

Retail SQL Analytics Product Pricing & Inventory Analysis using PostgreSQL

A structured business case study demonstrating data-driven decision-making through SQL-powered retail analytics.

ISHIKA JAIN

SQL BUSINESS ANALYTICS CASE STUDY



Business Context & Project Scope



Why This Analysis Matters

Retail businesses manage thousands of products with varying prices, discounts, and inventory levels. This project uses PostgreSQL to analyze a retail product dataset and answer business-focused questions on pricing strategy, inventory management, category performance, and revenue estimation.

Pricing Strategy

Optimize margins through discount and MRP analysis

Inventory Optimization

Monitor stock levels to prevent lost sales

Category Performance

Evaluate product segments for strategic decisions

PROBLEM STATEMENT

Business Problem & Project Objectives

Retail stakeholders need actionable insights to improve pricing decisions, monitor stock availability, evaluate discount strategies, and identify high-performing product categories.



Analyze Pricing Strategy

Evaluate MRP, selling price, and discount structures across products



Evaluate Discount Effectiveness

Identify which products and categories offer the most competitive pricing



Monitor Inventory Availability

Flag out-of-stock items, especially premium products with revenue impact



Compare Category Performance

Rank categories by estimated revenue, average discount, and inventory weight

Dataset Overview & Data Preparation

Zepto Product Dataset

Granularity: Product-Level Data

Field	Description
Product Name	Unique product identifier
Category	Product classification
MRP / Selling Price	Pricing fields
Discount %	Promotional depth
Available Qty	Stock on hand
Weight / Stock Status	Unit size & availability

Data Preparation Pipeline

Before analysis, the dataset was cleaned and validated to ensure reliable SQL outputs.

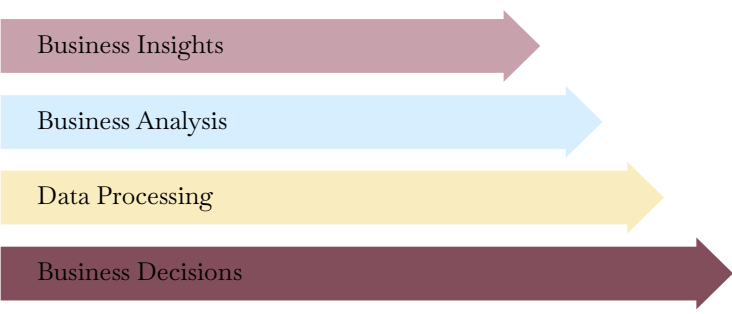


Each stage ensures data integrity before business analysis begins

SQL Concepts and Methodology

SQL Concepts Applied

- Filtering
- GROUP BY
- HAVING
- Aggregates
- CASE Statements
- ORDER BY



This end-to-end workflow transforms raw retail data into actionable business decisions through structured SQL analysis.

Business Questions Addressed

Top revenue categories?

Revenue

Question	Focus Area
Highest discount products?	Pricing
Premium products out of stock?	Inventory
Premium products with low discounts?	Pricing
Categories with highest avg. discount?	Discounts
Best value by price per gram?	Value
Products by package size?	Segmentation

FINDINGS

Key Business Insights

Highest Discount Products Identified

Top-discount SKUs surfaced for promotional benchmarking

Premium Out-of-Stock Products Flagged

High-value unavailable items represent recoverable revenue

Category Revenue Estimated

Revenue ranked by category to guide investment priorities

Price-Per-Gram Value Analysis

Unit economics reveal customer value differences across SKUs

Discount Variation Across Categories

Average discount differs significantly by category

Inventory Weight by Category

Weight distribution informs logistics and storage planning

Business Recommendations



Optimize Promotional Campaigns

Leverage high-discount products to drive traffic and conversion



Prioritize Premium Replenishment

Restock out-of-stock premium items to recover lost revenue



Review Premium Pricing Strategy

Reassess low-discount premium products for competitiveness



Improve Category Merchandising

Align shelf placement and promotions with category performance data



Benchmark Using Price-Per-Unit Metrics

Standardize value comparisons across package sizes and categories



Enhance Inventory Planning

Use category-level weight and stock data for demand forecasting

Business Impact & Skills Demonstrated

Business Impact Areas

- Pricing Optimization
- Inventory Management
- Category Performance Monitoring
- Merchandising Strategy
- Operational Planning
- Data-Driven Decision Making

Skills Matrix

Business Analysis	SQL	Retail Analytics
Problem Definition	Data Cleaning	Pricing Analysis
Question Formulation	Aggregations	Inventory Analytics
Recommendations	Filtering & GROUP BY	Product Performance
	HAVING & CASE	Category Analysis
	ORDER BY	

Project Architecture & Repository Overview

End-to-End Project Workflow

Business Problem

Retail Dataset

Data Cleaning

SQL Business Analysis

Business Insights

Business Recommendations

Business Decisions

Repository Structure

```
retail-sql-analytics/  
├── dataset/  
│   └── zepto_products.csv  
├── sql/  
│   └── retail_business_analysis.sql  
├── screenshots/  
│   └── query_outputs/  
├── report/  
│   └── business_case_study.pdf  
├── README.md  
└── LICENSE
```

CONCLUSION

Conclusion & Next Steps

This project demonstrates how PostgreSQL can be used to analyze product pricing, inventory availability, and category performance in a retail environment. The analysis supports data-driven decision-making for pricing optimization, stock management, and category strategy.

Future Enhancements

Sales Trend Analysis

Customer Segmentation

Dashboard Development

Tools Used

PostgreSQL

SQL

Business Analytics